

VENKAT AKHILA REDDY

AI/ML ENGINEER

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EXPERIENCE

AI Expert Contributor

Jul 2026 – Aug 2026

Marathon TS – Contract

Remote, USA

- Evaluated **Python and ML pull requests** for correctness, instruction adherence, numerical behavior, and edge-case handling in AI-assisted code evaluation workflows.
- Reviewed **NumPy and JAX** implementations against automated tests and technical specifications, identifying logical, numerical, and implementation-level errors.

LLM Researcher / ML Engineer

Jan 2025 – Jun 2026

University of Michigan–Dearborn

Dearborn, MI

- Fine-tuned and evaluated **LLaMA, Mistral, Qwen, and OpenChat** using **LoRA/PEFT** for narrative scoring across **1,900+ samples**, achieving expert-level reliability with **ICC 0.81–0.91**.
- Built scalable training, batch inference, evaluation, and endpoint deployment workflows on **GCP Vertex AI**; implemented **MLflow** experiment tracking, CI/CD evaluation, and structured model validation.
- Developed multi-model benchmarking and error-analysis pipelines, reducing critical prediction errors to **<2%**; monitored inference and compute behavior using **Prometheus and Grafana**.

AI Engineer Intern

Dec 2023 – Dec 2024

Product Manager Accelerator

Remote, USA

- Developed transformer-based document intelligence pipelines using **BERT, SciBERT, BioBERT, and Longformer** to classify and extract information from **1,000+ research documents**.
- Engineered automated PDF-processing workflows using **GROBID**, reducing manual screening workload by approximately **68%**; built **Python/FastAPI** inference services and CI/CD validation workflows.
- Curated and validated **1,000+ labeled samples** and evaluated models using precision, recall, F1, and systematic error analysis.

Data Scientist

Jan 2023 – Nov 2023

Gear Head Motors

Hyderabad, India

- Built **XGBoost and Random Forest** models for customer churn and demand forecasting, contributing to a **15% improvement in customer retention**.
- Developed **Python, SQL, and Databricks ETL pipelines** for model training and analytics; engineered lag and rolling time-series features and batch inference validation workflows.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Databricks, ETL, Apache Spark, Git/GitHub

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Random Forest, Model Evaluation

LLMs & GenAI: Hugging Face Transformers, LLaMA, Mistral, Qwen, LoRA/PEFT, RAG, LLM Evaluation

Agents & Retrieval: LangChain, LangGraph, Tool Calling, FAISS, BM25, Pinecone, Hybrid Retrieval

Cloud & MLOps: GCP, Vertex AI, MLflow, Docker, Kubernetes, CI/CD, Prometheus, Grafana

APIs & Serving: FastAPI, REST APIs, SSE Streaming, Batch Inference

PROJECTS

Agentic MLOps Analyst

Python, LangChain, MLflow, FastAPI

- Built an **LLM agent with custom tools** to query MLflow experiments, compare model metrics and hyperparameters, and generate recommendations grounded in deterministic experiment results.

SkinScope AI – Hybrid RAG Assistant

LangGraph, FAISS, BM25

- Built an agentic RAG system combining **FAISS + BM25 hybrid retrieval** with an **Answer → Critic → Rewrite** workflow and evidence-grounded responses.

EDUCATION

University of Michigan–Dearborn

Dec 2025

M.S. in Computer and Information Science

Dearborn, MI

Sreenidhi Institute of Science and Technology

Jun 2023

B.Tech. in Electronics and Computer Engineering

Hyderabad, India